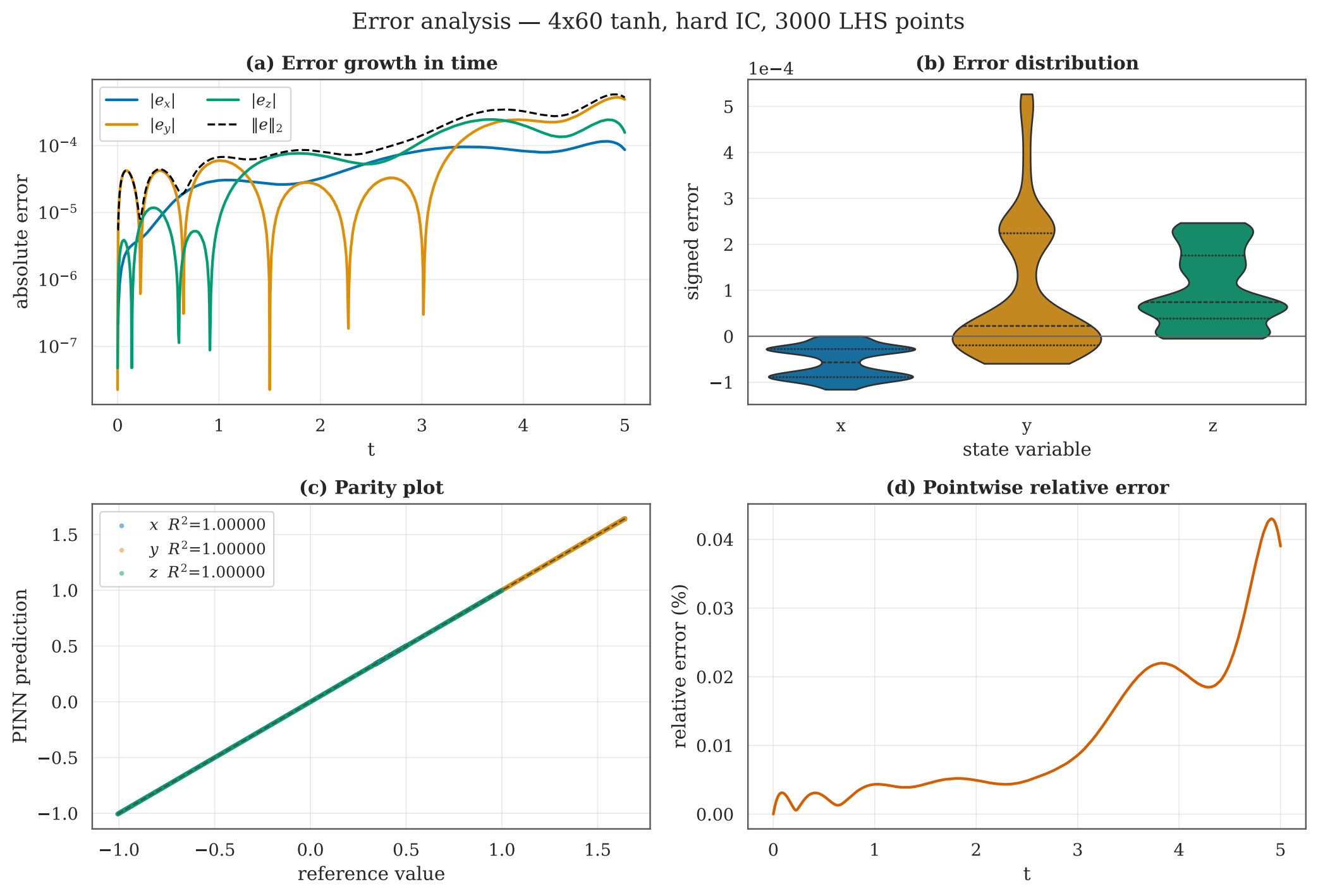

Error analysis — 4x60 tanh, hard IC, 3000 LHS points
(a) Error growth in time
$|e_x|$
$|e_y|$
$|e_z|$
$\|e\|_2$
absolute error
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
1
2
3
4
5
t
(b) Error distribution
1e−4
signed error
5
4
3
2
1
0
−1
x
y
z
state variable
(c) Parity plot
x R^2=1.00000
y R^2=1.00000
z R^2=1.00000
PINN prediction
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
reference value
(d) Pointwise relative error
relative error (%)
0.04
0.03
0.02
0.01
0.00
0
1
2
3
4
5
t